



RESCUEHACKS · EMERGENCY
RESPONSE

SKY EYE

WHAT IT IS

A triage accelerant for missing-person search. Ranked leads. A human verifies.

WHAT IT IS NOT

Not a finder. Not a confirmation.
Not a satellite detector.

WHERE WE ARE

Live console: report → ring → tiled
detect on drone photographs.

SkyEye surfaces possible leads only. It does not confirm a person's location or safety. Contact 911 / local SAR immediately.

NEXT STEPS, FUTURE DIRECTION & POTENTIAL IMPACT

SkyEye compresses the first hours of a search: unstructured report to a **Lost Person Behavior ring**, drone photograph to **ranked person-shaped candidates**. This note is the post-demo path — what to build next, who it is for, and how impact is measured without claiming a find.

02 WHERE WE STAND

The two loops that already meet on the map, and the measured limits of the current detector.

03 WHY THE HOURS MATTER

SAR is time-critical. Drone footage already exists. Review still happens by hand.

04 RECOMMENDED SEQUENCE

Ninety days, twelve months, eighteen months — in the order a field pilot can absorb.

05 DETECTION THAT MATCHES THE ALTITUDE

Aerial fine-tune, video ingest, thermal. The three upgrades that change recall.

06 PILOT, GUARDRAILS, ASK

One volunteer chapter. Historical flights first. No operational use until numbers exist.

HOURS OF SEARCHING. MINUTES TO A LEAD.

SECTION 01

WHERE WE STAND

The product already does the honest version of the pitch. **Where to look** is a map problem. **What to look at** is a drone-altitude photograph. Commercial satellite imagery tops out around 30–50 cm per pixel — a person is one or two pixels. Those tiles never enter the detector.

01 Report plain language	02 Extract Gemini / Groq	03 Geocode last-known	04 LPB ring 50th percentile	05 Photo drone nadir	06 Tiled YOLO ONNX · merge	07 Leads human reviews
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Loop A – where

REPORT → SEARCH AREA

Free-text intake. Structured facts: last-known place, elapsed hours, clothing, subject category. Google Geocoding. A simplified Koester 50th-percentile radius:
 $r = \text{clamp}(r_{50} \sqrt{t / t_0}, 200 \text{ m}, 8000 \text{ m}) \cdot t_0 = 3 \text{ h}$
 $\sqrt{t}$ keeps the ring from doubling when time doubles. An elderly hiker at 4.5 h gets $r \approx 1.3 \text{ km}$. Heuristic, not an operational plan.

Loop B – what

PHOTOGRAPH → RANKED CANDIDATES

Sliding-window YOLOv8n (ONNX Runtime).
Overlapping tiles, letterboxed to native input, projected back, merged across seams. Ranked boxes and map pins when the fixture is georeferenced.

Time-to-first-lead on laptop CPU: about 5–8 s on the slowest demo fixture, inside the 30 s target. UI never says “found.” Zero results are never “the area is clear.”

LIMIT	WHAT WE MEASURED	WHAT IT IMPLIES NEXT
Scale floor	Pretrained YOLOv8n misses subjects ≤ 60 px entirely. ≥ 100 px is reliable. Dropping confidence to 0.05 recovered none of them and added nine false positives. HERIDAL-class people sit at ~ 30 – 60 px.	Fine-tune on aerial SAR stills. Tiling cannot invent a prior the backbone never saw.
Pose prior	<code>lone_surfer_shorebreak</code> never detects at any threshold — prone, nadir, COCO's upright-person prior. Kept in the corpus on purpose.	Aerial labels (prone, partial, occluded) are the fix. Not a lower threshold.
Tiling tradeoff	<code>TILE_SIZE = 320</code> lifted a controlled scale test from 4/10 to 7/10 recall, at $\sim 2.5\times$ latency and $\sim 5\times$ false positives.	Recall lever for demo; a fine-tuned model should need less aggressive tiling.
True negatives	7 of 8 demo fixtures behave as expected. Both canopy true-negatives return zero false positives.	Empty-state copy stays. A future model must not “find” trees.

SECTION 02

WHY THE HOURS MATTER

Impact is not “AI finds missing people.” Impact is a **shorter time-to-first-verifiable-lead** and fewer reviewer-hours spent on empty canopy. Survival in land search declines with duration; planning quality in the first operational period is the difference between a ring that is searchable and one that has already doubled.

51 h

Search duration was the strongest survival predictor in a CART analysis of SAR incidents. A 51-hour mark captured nearly all survivors in the derivation set.

>90%

Wilderness searches resolved by early reflex tasks. The remainder reward intensive planning — probability maps, effort allocation, updates after each period.

227 h

Person-hours spent reviewing 8,399 UAS photographs in one published missing-person case, across 53 reviewers and thousands of hectares.

Adams et al., *Wilderness & Environmental Medicine* (2007), “Search Is a Time-Critical Event.” Sava et al., *Transactions in GIS* (2015), evaluating Lost Person Behavior maps; Koester (2008) / ISRID. Díez-González et al., high-resolution aerial photography for missing persons (DOI 10.6036/10631).

Mechanism 1

A RING FROM EVIDENCE

Dispatch still gets a scared caller and a guessed radius. SkyEye extracts a subject category and elapsed time, then sizes a 50th-percentile LPB ring. That does not replace an incident commander. It gives the first map something other than a round number.

Mechanism 2

PIXELS, RANKED

Volunteer chapters already own drones. The bottleneck is eyes on stills and video. SkyEye does not make a person more present in a frame. It spends a finite attention budget on the most person-shaped pixels first.

Mechanism 3

UNCERTAINTY, KEPT

False certainty kills trust and can pull ground teams to the wrong draw. Ranked scores, a non-dismissible emergency banner, and empty-state copy that refuses “all clear” are part of the impact model, not decoration.

Who this is for. Near term: a volunteer SAR coordinator triaging drone stills, or a dispatcher who already called 911 and needs a defensible first ring. Later: mountain, coastal, and wilderness chapters that fly UAS today and still review footage by hand. Not the bystander as a freelance searcher. The product never tells anyone to walk toward a pin.

We will not report “lives saved.” A responsible scorecard is: time from report (or landing) to first reviewed lead; precision/recall on that chapter’s own imagery; reviewer-hours per km² of coverage; and whether night/thermal extends the window when most searches go dark. Satellite GSD does not move. Detection stays on drone photographs.

SECTION 03

RECOMMENDED SEQUENCE

Build in the order a field chapter can absorb. Do not ship live ingest before the model has seen nadir people, and do not put SkyEye on an active incident before it has been scored on that chapter's historical flights.

90 DAYS

STABILIZE ·
INVITE

MAKE THE DEMO A TOOL A COORDINATOR COULD SIT WITH

- **Production detect.** Finish the ONNX Runtime path on Render (no PyTorch in the running image). One Unicorn worker. Operator Docker rebuild — not a restart. Confirm `model.weights = yolov8n.onnx`.
- **Regression harness.** Keep the eight-fixture corpus. Treat the 60 px floor and the surfer miss as tests a future weight file must beat without lighting up the canopy true-negatives.
- **One chapter conversation.** Volunteer SAR that already flies. Tabletop only: how they capture, how they review, whether geotags exist. No operational use.
- **Dataset access.** Request HERIDAL and/or SARD (gated). Fine-tune cannot start on Wikimedia substitutes alone.
- **Capture note.** One page for pilots: nadir, overlap, geotags, GSD target so a person is large enough to be a detection problem rather than a physics problem.

12 MONTHS

DETECTION ·
INGEST

MATCH THE ALTITUDE SAR ACTUALLY FLIES

- **Aerial fine-tune** on HERIDAL/SARD so prone, partial, 30–60 px subjects are in-distribution. Hold out a slice; quote a real recall number.
- **Video stills.** Sample frames from footage teams already record. Same `/api/detect` contract. Cluster in time so one person is not twenty pins.
- **Thermal / IR** only if the partner already carries the payload. Highest leverage for night search; different false-positive profile (rock, livestock, residual heat).

18 MONTHS

INTO THE
INCIDENT

ONLY AFTER NUMBERS ON THEIR IMAGERY

- **Limited operational trial** on a non-critical training search or after-action review — never as the sole cue for ground teams.
- **Airframe ingest** (DJI SDK / MAVLink) instead of file upload, if the chapter wants it.
- **Multi-report clustering** — several witnesses, one merged ring. Stretch in the original PRD; useful once intake is trusted.
- **Field / offline mode** for trailheads where the cloud is a joke. Same safety copy, local ONNX.

What stays out until the sequence says so

SATELLITE DETECT · GLOBE-AS-PRODUCT · “FOUND” UX

Running YOLO on a basemap will fail a GIS-literate review in thirty seconds. A decorative globe on the landing page is not a search surface. Changing the frozen `/api/detect` shape mid-pilot is how the console and the model stop talking. Those are strategy, not backlog shame.

SECTION 04

DETECTION THAT MATCHES THE ALTITUDE

Pretrained COCO is a street-level, upright-person prior. Wilderness SAR is nadir, small, often prone, often partial. The next model work is not a bigger prompt. It is labels that look like the job.

Upgrade A

AERIAL FINE-TUNE

HERIDAL, SARD, optionally C2A. Purpose-built drone-altitude boxes. Train so the 60 px floor moves down into the 30–60 px band where those datasets live.

Done looks like: held-out recall quoted in the README; surfer-class poses no longer a permanent miss; canopy true-negatives still clean; tiling less aggressive for the same recall.

Upgrade B

VIDEO INGEST

Teams do not hand you one JPEG. They land with a card of 4K. Sample ~1 fps (or keyframes), run the existing detector, stitch boxes that persist across frames, drop flicker.

Done looks like: an operator can drop an MP4, get a timeline of leads, jump to the frame, see the same safety banner. Not a live siren from the aircraft.

Upgrade C

THERMAL / IR

The original PRD called this the highest-impact real-world upgrade: most searches continue after dark, when EO stills go useless. Thermal is a different sensor and a different error mode.

Done looks like: a second weight file, dual-mode UI, calibrated scores, and a partner who already owns the payload. Do not buy a camera to demo a story.

HOW WE WILL REFUSE TO OVERFIT THE DEMO

The Wikimedia corpus is openly licensed and the right kind of hard — people a few dozen pixels across canopy — but it is not HERIDAL. Demo-placed pins on the Bruce Trail / Milton conservation area are labeled as such. A partner trial uses *their* geotagged stills, or it does not project to the map.

Fine-tune success is not “more boxes on the beach fixture.” It is recall on held-out aerial people *and* an unchanged empty canopy. If precision collapses, we did not ship a rescue tool; we shipped a hallucination with GPS.

ENGINEERING THAT MUST NOT REGRESS

Render’s 2 GB class killed a full PyTorch stack (status 137). Runtime stays ONNX, one worker, tight threads. Weights bake into the image. The browser talks to Render directly — Vercel Hobby will time out tiled CPU detect.

`/api/detect` stays a ranked list of boxes, scores, optional lat/long. Video is extra frames of the same object. Thermal is another model behind the same shape. Keys stay off the wire and out of logs. Rate limits stay on extract, geocode, and detect.

CRASAR-shaped lesson. Wide-area drone search already produces tens of gigabytes that humans cannot finish. Computer vision on that pile is useful only when flights are nadir, overlapping, and geotagged for post-processing — not ad-hoc video shot “in case.” SkyEye’s capture note exists so a chapter does not collect the wrong data and then blame the model.

SECTION 05

PILOT, GUARDRAILS, ASK

The post-hackathon path in the PRD is unchanged: a partnership with a volunteer SAR chapter that already owns drones and still reviews footage manually. One chapter. Historical flights first. A human always between SkyEye and the ground team.

Pilot design
WHAT WE RUN

- De-identified stills or sampled video from a closed search or a training exercise.
- Blind review: coordinator marks leads they would have flagged by hand; we compare time-to-first-lead and missed persons in-frame.
- Map pins only where EXIF/flight logs support them. No demo-placement in a trial write-up.
- Stop rule: if false positives bury the list, we do not "tune the demo" — we report the miss and go back to weights or tiling.

Scorecard
WHAT WE PUBLISH

- Recall / precision on that chapter's holdout, with the 60 px and pose cases called out.
- Seconds from ingest to first reviewed candidate vs. their current practice.
- Reviewer-minutes per km² before and after ranking.
- A written statement that SkyEye was not used to clear ground or to withhold a search.

Guardrails that do not expire

LANGUAGE

No "found," "located," or "confirmed" in the product. Confidence is always visible. Empty is not clear.

ACTION

No feature suggests a bystander walk to a pin. Emergency / local SAR contact stays on every result screen.

SENSOR

No detector on map or satellite tiles. GSD is physics. The workaround is altitude, not marketing.

DATA

No live surveillance corpus. Partner imagery stays theirs. Keys never in git, responses, or logs.

The ask
ONE CHAPTER · HISTORICAL FLIGHTS · A REVIEWER

Access to HERIDAL or SARD so the model's prior matches nadir SAR. A volunteer chapter willing to sit through a tabletop, then a scored replay of flights they already flew. Permission to quote aggregate metrics, not subjects' names. After that, and only after that: a conversation about video, thermal, and whether SkyEye belongs on the next training search.

**01
This week**

Production ONNX on Render. Leave extract, geocode, and the map contract as they are.

**02
This quarter**

HERIDAL/SARD request, one SAR tabletop, and a one-page capture protocol.

**03
This year**

Fine-tune with a published holdout number. Video stills on partner footage.

**04
After the number**

Thermal if they carry the payload. Airframe ingest if they want it. Never "all clear."